GLOBAL ECONOMICS

10th grade

Term 2 – Lesson 3 MATERIAL: Decision Analysis with Bayes, Maximin, and Minimax Regret (Worked Solutions)

C8: Comparison of Bayes, Maximin, and Minimax Regret on the same dataset

1.

Problem description

An export cooperative must choose one production plan for next month: Plan A_1, Plan A_2, or Plan A_3. It models three demand states with probabilities:

$$S_1 : 0.40,\ S_2 : 0.35,\ S_3 : 0.25,$$

which sum to 1.00. Payoffs are in thousand USD. The values reflect that A_3 earns most in a boom because it uses overtime capacity, while A_2 protects profits better when demand softens thanks to lower fixed costs. Apply Bayes (EV), Maximin, and Minimax Regret.

C8

Step 1. Build the payoff table.

Payoff table

State of nature	Probability	A_1	A_2	A_3
S_1	0.40	140	130	155
S_2	0.35	95	105	90
S_3	0.25	60	80	45

Step 2. Bayes criterion (Expected Value) using $EV(A_i) = \sum p_j \cdot x_{ij}$.

$$EV(A_1) = 0.40(140) + 0.35(95) + 0.25(60) = 104.25,$$
$$EV(A_2) = 0.40(130) + 0.35(105) + 0.25(80) = 108.75,$$
$$EV(A_3) = 0.40(155) + 0.35(90) + 0.25(45) = 104.75.$$

EV ranking (best → worst):

$$A_2\,(108.75) > A_3\,(104.75) > A_1\,(104.25).$$

Step 3. Maximin criterion.

$$\min(A_1) = 60, \quad \min(A_2) = 80, \quad \min(A_3) = 45.$$

Maximin ranking (best → worst):

$$A_2\,(80) > A_1\,(60) > A_3\,(45).$$

Step 4. Minimax Regret criterion. Find the best payoff in each state:

$$\max(S_1) = 155, \quad \max(S_2) = 105, \quad \max(S_3) = 80.$$

Build regret table using $r_{ij} = \max(S_j) - x_{ij}$.

Alternative	S_1	S_2	S_3	Max regret
A_1	$155 - 140 = 15$	$105 - 95 = 10$	$80 - 60 = 20$	20
A_2	$155 - 130 = 25$	$105 - 105 = 0$	$80 - 80 = 0$	25
A_3	$155 - 155 = 0$	$105 - 90 = 15$	$80 - 45 = 35$	35

Minimax Regret ranking (best $\rightarrow$ worst, lower is better):

$$A_1\,(20) < A_2\,(25) < A_3\,(35).$$

Step 5. Conclusion. Bayes and Maximin both select A_2, while Minimax Regret selects A_1. The rankings show how a safer regret profile can differ from the highest expected payoff.

2.

Problem description

A transport company must choose one fleet plan among six alternatives: $B_1, B_2, B_3, B_4, B_5, B_6$. There are three fuel-demand states with probabilities:

$$T_1 : 0.50, \quad T_2 : 0.30, \quad T_3 : 0.20,$$

and $0.50 + 0.30 + 0.20 = 1.00$. Payoffs are in thousand USD. Apply Bayes (EV), Maximin, and Minimax Regret.

C8

Step 1. Build the payoff table.

Payoff table

State of nature	Probability	B_1	B_2	B_3	B_4	B_5	B_6
T_1	0.50	95	90	110	85	100	92
T_2	0.30	70	75	60	80	65	78
T_3	0.20	30	40	10	55	35	20

Step 2. Bayes criterion (Expected Value).

$$\begin{aligned}
EV(B_1) &= 0.50(95) + 0.30(70) + 0.20(30) = 74.50, \\
EV(B_2) &= 0.50(90) + 0.30(75) + 0.20(40) = 75.50, \\
EV(B_3) &= 0.50(110) + 0.30(60) + 0.20(10) = 75.00, \\
EV(B_4) &= 0.50(85) + 0.30(80) + 0.20(55) = 77.50, \\
EV(B_5) &= 0.50(100) + 0.30(65) + 0.20(35) = 76.50, \\
EV(B_6) &= 0.50(92) + 0.30(78) + 0.20(20) = 73.40.
\end{aligned}$$

EV ranking (best $\rightarrow$ worst):

$$B_4\,(77.50) > B_5\,(76.50) > B_2\,(75.50) > B_3\,(75.00) > B_1\,(74.50) > B_6\,(73.40).$$

Step 3. Maximin criterion.

$$\begin{aligned}
&\min(B_1) = 30, \ \min(B_2) = 40, \ \min(B_3) = 10, \\
&\min(B_4) = 55, \ \min(B_5) = 35, \ \min(B_6) = 20.
\end{aligned}$$

Maximin ranking (best $\rightarrow$ worst):

$$B_4\,(55) > B_2\,(40) > B_5\,(35) > B_1\,(30) > B_6\,(20) > B_3\,(10).$$

Step 4. Minimax Regret criterion. Best payoff in each state:

$$\max(T_1) = 110, \quad \max(T_2) = 80, \quad \max(T_3) = 55.$$

Regret table:

Alternative	T_1	T_2	T_3	Max regret
B_1	$110 - 95 = 15$	$80 - 70 = 10$	$55 - 30 = 25$	25
B_2	$110 - 90 = 20$	$80 - 75 = 5$	$55 - 40 = 15$	20
B_3	$110 - 110 = 0$	$80 - 60 = 20$	$55 - 10 = 45$	45
B_4	$110 - 85 = 25$	$80 - 80 = 0$	$55 - 55 = 0$	25
B_5	$110 - 100 = 10$	$80 - 65 = 15$	$55 - 35 = 20$	20
B_6	$110 - 92 = 18$	$80 - 78 = 2$	$55 - 20 = 35$	35

Minimax Regret ranking (best $\rightarrow$ worst, lower is better):

$$B_2\,(20) = B_5\,(20) > B_1\,(25) = B_4\,(25) > B_6\,(35) > B_3\,(45).$$

Ties are explicit between (B_2, B_5) and between (B_1, B_4).

Step 5. Conclusion. The three criteria lead to different priority orders, so decision choice depends on risk attitude.

3.

Problem description

A cocoa processor must choose one export contract strategy: C_1, C_2, or C_3. Three exchange-rate states have probabilities $R_1 : 0.35,\ R_2 : 0.40,\ R_3 : 0.25$.

Payoffs are in thousand USD. C_3 has the highest upside if the currency weakens, but C_2 includes hedge clauses that stabilize returns in moderate and strong-currency conditions.

State of nature	Probability	C_1	C_2	C_3
R_1	0.35	135	125	150
R_2	0.40	100	110	92
R_3	0.25	70	85	50

C8

Step 1. Use the payoff table.

Step 2. Bayes criterion.

$$EV(C_1) = 104.75, \quad EV(C_2) = 108.75, \quad EV(C_3) = 104.30.$$

EV ranking: $C_2 > C_1 > C_3$.

Step 3. Maximin criterion.

$$\min(C_1) = 70, \quad \min(C_2) = 85, \quad \min(C_3) = 50.$$

Maximin ranking: $C_2 > C_1 > C_3$.

Step 4. Minimax Regret criterion.

Alternative	R_1	R_2	R_3	Max regret
C_1	15	10	15	15
C_2	25	0	0	25
C_3	0	18	35	35

Minimax Regret ranking: $C_1 < C_2 < C_3$ (lower is better).

Step 5. Conclusion: Bayes and Maximin choose C_2, while Minimax Regret favors C_1.

4.

Problem description

A digital bank evaluates six lending policies D_1 to D_6 under three default-rate states V_1, V_2, V_3 with probabilities $0.45, 0.35, 0.20$.

Payoffs are in thousand USD. Aggressive policies perform better when defaults stay low, while conservative policies lose less in the weak state due to tighter credit scoring.

State of nature	Probability	D_1	D_2	D_3	D_4	D_5	D_6
V_1	0.45	102	95	118	90	108	98
V_2	0.35	74	82	66	86	72	80
V_3	0.20	28	44	8	58	36	24

C8

Step 1. Bayes EV values:

$$EV(D_1) = 77.80,\ EV(D_2) = 82.85,\ EV(D_3) = 78.40,\ EV(D_4) = 84.80,\ EV(D_5) = 81.90,\ EV(D_6) = 76.70.$$

EV ranking: $D_4 > D_2 > D_5 > D_3 > D_1 > D_6$.

Step 2. Maximin values:

$$\min(D_1) = 28,\ \min(D_2) = 44,\ \min(D_3) = 8,\ \min(D_4) = 58,\ \min(D_5) = 36,\ \min(D_6) = 24.$$

Maximin ranking: $D_4 > D_2 > D_5 > D_1 > D_6 > D_3$.

Step 3. Minimax Regret (state bests $118, 86, 58$) gives max regrets $D_1 : 30, D_2 : 23, D_3 : 50, D_4 : 28, D_5 : 22, D_6 : 34$.

Minimax Regret ranking: $D_5 < D_2 < D_4 < D_1 < D_6 < D_3$ (lower is better).

Step 4. Conclusion: Bayes and Maximin choose D_4, while Minimax Regret selects D_5.